

SRI ESHWAR COLLEGE OF ENGINEERING

MODEL QUESTION PAPER

Subject : MERN

Time : 3 Hours

Max Marks : 100

PART - A

1. Which of the following protocols is primarily used for securely transferring web pages over the internet? (1 Mark)

- A. HTTP
- B. FTP
- C. HTTPS
- D. SMTP

2. What does JSX stand for in React? (1 Mark)

- A. JavaScript XML
- B. JavaScript Extension
- C. JSON XML
- D. JQuery Syntax Extension

3. Which MongoDB data type is suitable for storing a unique identifier that can be used across multiple machines without coordination? (1 Mark)

- A. String
- B. NumberInt
- C. ObjectID
- D. Boolean

4. Which React Hook is primarily used for managing state in functional components? (1 Mark)

- A. useEffect
- B. useContext
- C. useState
- D. useReducer

5. Which command is used to install a package as a development dependency in Node.js? (1 Mark)

- A. npm install --save
- B. npm install --global
- C. npm install --save-dev
- D. npm update

6. What is a callback function in JavaScript? (1 Mark)

- A. A function that returns another function
- B. A function passed as an argument to another function
- C. A function that executes immediately upon definition
- D. A function used for error handling only

7. Node.js is built on which JavaScript engine? (1 Mark)

- A. SpiderMonkey
- B. V8
- C. JavaScriptCore
- D. Rhino

8. Which Flexbox property specifies the alignment of items along the main axis? (1 Mark)

- A. align-items
- B. justify-content
- C. flex-direction
- D. align-self

9. Variables declared with `const` in JavaScript can be reassigned a new value after their initial assignment. (1 Mark)

10. By default, the `useEffect` hook in React functional components runs only once after the initial render. (1 Mark)

11. MongoDB is a relational database management system that uses SQL for querying data. (1 Mark)

12. In Node.js, `module.exports` is used to define what a module exports and makes available to other modules. (1 Mark)

13. CSS Grid is primarily designed for one-dimensional layout content. (1 Mark)

14. An `async` function in JavaScript always returns a Promise. (1 Mark)

15. The default port for HTTP communication is _____. (1 Mark)

16. A React component typically returns a single _____ element. (1 Mark)

17. The `\_\_\_\_\_` middleware is commonly used in Express.js to parse incoming request bodies in JSON format. (1 Mark)

18. A JavaScript function that calls itself until a base condition is met is known as a _____ function. (1 Mark)

19. Mongoose provides an Object Data Modeling (ODM) solution for _____ databases in Node.js applications. (1 Mark)

20. The `\_\_\_\_\_` hook is used in React functional components to perform side effects like data fetching or DOM manipulation. (1 Mark)

PART - B

21. Differentiate between `let` and `var` keywords for variable declaration in JavaScript. (2 Marks)
22. Explain the significance of the `key` prop when rendering lists of components in React. (2 Marks)
23. What is the role of middleware in an Express.js application? Give a basic example. (2 Marks)
24. Define Responsive Web Design and state its primary goal. (2 Marks)
25. Briefly explain the fundamental difference in data storage philosophy between SQL and NoSQL databases. (2 Marks)

PART - C

26.

- a) Explain the concepts of 'State' and 'Props' in React components. Discuss their differences, typical use cases, and how they contribute to dynamic user interfaces with suitable code examples. (10 Marks)
- b) Describe the purpose of the `useState` hook in functional React components. (4 Marks)

27.

- a) Elaborate on the 'Callback Hell' problem encountered in traditional asynchronous JavaScript programming. How do JavaScript Promises provide a cleaner and more manageable approach to handle asynchronous operations? Illustrate with a simple code example using Promises. (10 Marks)
- b) How does the `async/await` syntax further improve readability and error handling compared to chaining Promises? (4 Marks)

28.

- a) Design a basic RESTful API structure using Express.js for a simple 'Tasks' management application. Your API should handle operations like creating a new task, retrieving all tasks, retrieving a specific task by ID, and deleting a task. Provide the Express.js route definitions and placeholder handler functions for each endpoint. (10 Marks)
- b) Explain the concept of route parameters in Express.js and provide an example of their use. (4 Marks)

29.

- a) Discuss the key characteristics and advantages of using a NoSQL database like MongoDB for modern web applications, particularly within the MERN stack. Explain the core concepts of 'Documents' and 'Collections' in MongoDB, and how Mongoose simplifies schema definition and data interaction in Node.js applications. (10 Marks)
- b) Create a Mongoose schema definition for a `BlogPost` model that includes fields for `title` (String, required), `content` (String), `author` (String), and `tags` (array of Strings). (4 Marks)

30.

- a) Compare and contrast CSS Flexbox and CSS Grid layout modules. Discuss their primary use cases, their fundamental differences (one-dimensional vs. two-dimensional), and provide scenarios where one might be more suitable than the other. Include brief CSS examples to illustrate their basic application. (10 Marks)
- b) What are CSS Custom Properties (CSS Variables) and how do they enhance maintainability and flexibility in styling large web projects? (4 Marks)

31.

- a) Provide a detailed architectural overview of a MERN stack application. Explain how MongoDB, Express.js, React, and Node.js interact with each other to form a cohesive full-stack solution, outlining the typical data flow from the client to the database and back. (10 Marks)
- b) Briefly explain how the MERN stack aligns with or deviates from the traditional Model-View-Controller (MVC) architectural pattern. (4 Marks)

32.

- a) Explain the concept of the 'Virtual DOM' in React. Discuss how React utilizes the Virtual DOM to efficiently update the user interface and improve performance compared to direct manipulation of the browser's native DOM. Illustrate the reconciliation process. (10 Marks)

b) What are 'Synthetic Events' in React, and how do they differ from native browser events? (4 Marks)